DATA ITEM DESCRIPTION

Title: SOFTWARE SYSTEMS PRODUCT DESIGN (SSPD)

Number: DI-SESS-82466
AMSC Number: N10526
DTIC Applicable: N/A
Preparing Activity: AS
Applicable Forms: N/A

Approval Date: 20250127
Limitation: N/A
GIDEP Applicable: N/A
Project Number: SESS-2025-009

Use/relationship: The Software Systems Product Design (SSPD) describes the system-wide design decisions, the architectural design, and the detailed design of the training system. The SSPD includes detailed description of the system being developed, procured, maintained, or reused that is generated by specific and discrete task requirements as delineated in the contract.

The SSPD is an engineering Data Item Description (DID) that provides the acquirer visibility into the design content, provides information needed for technical support, and forms the basis for further training system development.

This DID contains the format, content, and intended use information for the data deliverable resulting from the work task described in the solicitation.

This DID supersedes DI-SESS-82036.

Requirements:

1. Referenced documents: Not applicable.

2. Format: The SSPD shall be in contractor format.

3. Content: The SSPD shall contain the following:

3.1.Title page: The SSPD shall have a title page providing the following information, as applicable:

3.1.1. Document identification numbering.

3.1.2. Volume number.

3.1.3. Version or revision indicator.

3.1.4. Security classification.

3.1.5. Release date.

3.1.6. Document title.

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3.1.7. Name of the training system to which this SSPD applies.

3.1.8. Name of each of the SIs to which this SSPD applies.

3.1.9. Contract number.

3.1.10. Contract Data Requirements List (CDRL) item Number.

3.1.11. Organization for which the document has been prepared.

3.1.12. Distribution statement.

3.1.13. Controlled Unclassified Information (CUI) .

3.1.14. Export-Control warning.

3.1.15. Destruction notice.

3.1.16. Record of changes.
The SSPD shall have a record of change from the previous document submission. Each submission shall append the changes to the record of changes section. The submissions shall have track changes turned on to record edits made in the document.

3.1.17. Table of contents.
The SSPD shall have a table of contents providing the number, title, and page number for each titled paragraph, figure, table, and appendix.

3.1.18. Page numbering.
Each page shall have a unique page number and include the document number, document version, volume, and date, as applicable.

3.1.19. Styles.
Diagrams, tables, and other presentation styles are acceptable substitutes for text when the data required by this DID can be made more readable using these styles.

3.1.20. Multiple paragraphs and subparagraphs.
Any section, paragraph, or subparagraph may be written as multiple paragraphs or subparagraphs to enhance readability.

3.1.21. Duplicated content
Any section, paragraph, or subparagraph shall include dissimilar information to avoid the duplication of information.

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3.1.22. Design conventions.
Design conventions used within the SSPD to convey the design information shall be described or referenced.

3.1.23. Substitution of existing documents.
Existing documents may be substituted for part of the SSPD provided those documents contain the required information as defined by this DID and do not convey additional distribution restrictions to the SSPD. The substitution of existing documents shall be included in the SSPD either as an appendix to supplement the SSPD or as a separate document.

3.2.24. Document topics and numbering.
The following paragraph titles and content shall be included in the SSPD.

3.2.Content Requirements**:**

3.2.1. Scope
This section shall be divided into the following paragraphs:

3.2.1.1. Identification.
This paragraph shall fully identify the training system to which this SSPD applies, including as applicable, identifying number(s), title(s), abbreviation(s), acronym(s), version number(s), release number(s), and other identifying information.

3.2.1.2. System overview.
This paragraph shall summarize the purpose of the training system to which this SSPD applies. It shall describe the general nature of the training system; summarize the training system development history; identify the project sponsor, acquirer, user(s), developer(s), and support agencies; identify current and planned operating sites; and list relevant documents.

3.2.1.3. Document overview.
This paragraph shall summarize the purpose of this SSPD and describe the security or privacy considerations associated with its use.

3.2.2. Referenced Material .
This section shall list the title, number, revision, and date of the documents referenced within the SSPD. This section shall identify the source(s) for the documents not available through normal Government procuring activities.

3.2.3. Hardware Product(s).
This section shall describe the physical environment in which the system must operate. This section shall include the physical resources and hardware components of each subsystem. This section shall identify a list of the planned hardware part numbers from the original equipment manufacturer (OEM) required to satisfy the design. The information for this section shall be divided into the following paragraphs and subparagraphs:

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3.2.3.1. Identify the hardware components of each subsystem. Each hardware component shall be assigned a project-unique identifier.

3.2.3.2. Hardware design decisions made in response to requirements, such as selected approach to providing required flexibility, availability, and maintainability.

3.2.3.3. Show bidirectional traceability between the hardware components and hardware requirements.

3.2.3.4. Show the relationship(s) of the hardware components. Multiple relationships may be presented, depending on the selected design methodology.

3.2.3.5. State the purpose of each hardware component and identify the requirements and system-wide design decisions allocated to it.

3.2.3.6. Identify each hardware component's development status (such as new development, existing hardware component to be reused as is, existing design to be reused as is, existing design or hardware component to be modified, hardware component to be developed for reuse, hardware component planned for Build X, etc.). For existing design or hardware components, the description shall provide identifying information, such as name, version, series, documentation references, location, etc.

3.2.3.7. For each hardware component identified for use in each of the subsystems, describe its hardware resources (i.e., the processors, memory, IO devices, data storage, and network equipment). Each description shall, as applicable, identify the configuration items that will use the resource, describe the allocation of resource utilization to each subsystem that will use the resource (e.g., 20% of the resource's capacity allocated to the subsystem 1, 30% to subsystem 2), describe the conditions under which utilization will be measured, and describe the characteristics of the resource:

3.2.3.7.1. Descriptions of computer processors shall include, as applicable, manufacturer name and model number, and processor speed.

3.2.3.7.2. Descriptions of memory shall include, as applicable, manufacturer name and model number and memory size, type, speed, and configuration (i.e., 256MB cache memory, 16GB RAM (4GB x 4)).

3.2.3.7.3. Descriptions of IO device(s) shall include, as applicable, manufacturer name and model number, type of device, and device speed and capacity.

3.2.3.7.4. Descriptions of storage shall include, as applicable, manufacturer name and model number, type of storage, amount of installed storage, and storage speed.

3.2.3.7.5. Descriptions of the network equipment, such as network interface cards, hubs, gateways, cabling, high speed data lines, or aggregates of these or other hardware components, shall include, as applicable, manufacturer name and model number, data transfer rates, capacities, network topologies, transmission techniques, and protocols used.

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3.2.3.7.6. Each description shall also include, as applicable, growth capabilities, diagnostic capabilities, and any additional hardware capabilities relevant to the description.

3.2.4. Tradeoff Analysis.
This section shall provide a tradeoff analysis that includes decisions that involve procurement costs, maintenance costs, diminishing or losing on quality, quantity, or property of a set or design in return for gains in other aspects. The section shall include different design concepts and decisions made on the selected designs.

3.2.5. Data Rights Assertion List Design Changes.
This section shall identify the rights in technical data, the rights in other than commercial computer software, and the rights in other than commercial computer software documentation as specified in DFARS clauses 252.227-7013 and 252.227-7014. This section shall identify and highlight any data rights assertion changes since contract award and the basis for the assertion and rationale for the assertion change.

3.2.6. SI-wide design decisions.
This section shall identify every SI and describe every SI-wide design decision. An SI is an aggregation of software that is designated for configuration management and treated as a single entity in the configuration management process and SI-wide design decisions are those that impact the design of one or more SIs. Supporting rationale shall be included for the SI-wide design decisions that are not explicitly driven by the specified requirements. SI-wide design decisions in response to the requirements shall be identified. The following SI-wide design decisions shall be included as applicable:

3.2.6.1. Decisions related to behavior(s) and action(s) in response to each input, state, or condition.

3.2.6.2. Decisions related to response time and other performance characteristics.

3.2.6.3. Decisions related to data rights.

3.2.6.4. Decisions that address critical requirements, such as safety, cybersecurity or privacy.

3.2.6.5. Decisions that depend upon, or are related to, specific training system states or modes.

3.2.6.6. Decisions regarding each SI’s interface with users.

3.2.6.7. Decisions regarding each SI’s inputs and outputs with other distributed training networks (or environment).

3.2.6.8. Decisions regarding each SI’s inputs and outputs with other SIs.

3.2.6.9. Decisions regarding each SI's inputs and outputs with hardware configuration items (HWCIs).

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3.2.6.10. Decisions related to the modeling of the physical platform operational equipment (OE).

3.2.6.11. Decisions related to the selection of equations, algorithms, and rules.

3.2.6.12. Decisions related to the probability of kill, damage assessment, or dead reckoning.

3.2.6.13. Decisions on the selection of an interoperability standard for distributed simulation.

3.2.6.14. Decisions related to the handling of improper, un-allowed, and incorrect inputs or conditions.

3.2.6.15. Decisions regarding how information, data, and databases appear to the user.

3.2.6.16. Decisions made in response to the requirements, such as stability, availability, and supportability.

3.2.6.17. Decisions related to programming language selection.

3.2.6.18. Decisions related to the reuse of previously constructed source code.

3.2.6.19. Decisions related to the implementation of Artificial Intelligence (AI).

3.2.6.20. Other SI-Wide design decisions.

3.2.7. Software Product(s).
This paragraph shall identify all commercial and noncommercial software, such as the operating system, hardware device drivers, virtualization technology, computer software, and computer databases for each computational device that stores or processes data. Release date and version number shall be listed for each software item.

3.2.8. SI architectural design.

3.2.8.1. If any part of the architectural design depends upon system states or modes, this dependency shall be indicated within the architectural design.

3.2.8.2. If architectural design information is duplicated in more than one paragraph, present the information once and reference the other paragraphs.

3.2.9. SI structure and hierarchy .

3.2.9.1. This paragraph shall identify the software components (SCs) and the software units (SUs) that comprise the SI for each of the SIs. An SU is defined as the lowest software element and an SC is collection of SUs.

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3.2.9.2. Each SU shall be assigned a unique project identifier. SCs may occur at different levels of a hierarchy within the SI, and the SSPD may refer to SUs by any naming convention that is consistent with the design methodology being used.

3.2.9.3. This paragraph shall show the relationship and hierarchy of the SUs that make up the SI. Multiple relationships may be presented, depending on the selected design methodology. For example, in an object-oriented design, this relationship may be presented using class diagrams and object structures as well as module and process architectures of the SI.

3.2.9.4. This paragraph shall state the purpose of each SU.

3.2.9.5. This paragraph shall identify each of the SI-wide design decisions applicable to each SU.

3.2.9.6. This paragraph shall identify the software requirement(s) allocated to each SU.

3.2.9.7. This paragraph shall identify each SUs development status, such as new, reused, and modified software. Reused and modified software shall have identifying information such as the original software name/identifier, vendor/source, version, supporting documentation references, etc.

3.2.9.8. This paragraph shall describe the allocation of the SI to HWCI(s).

3.2.9.9. This paragraph shall describe each of the SI's planned utilization of computer hardware resources, such a processor capacity, memory capacity, disk storage capacity, input/output device capacity, and network capacity.

3.2.9.9.1. This description shall cover the spare resourcing requirements affecting the SI.

3.2.9.9.2. The assumptions and conditions on which the utilization data are based shall be described, such as during typical usage, worst-case usage, and specific conditions.

3.2.9.9.3. Special considerations affecting utilization, such as multiprocessors and the use of virtual memory, shall be described.

3.2.9.9.4. The utilization shall be described with units of measure, such as a percent of processor capacity, cycles per second, gigabytes of memory, and kilobytes per second.

3.2.10. Concept of execution
This section shall describe the concept of execution among the SUs comprising of the SI for each SI. Diagrams and descriptions shall show the dynamic relationship of the SUs during the operation of the SI. Descriptions of execution flow control, data flow, dynamically controlled sequencing, state transition diagrams, priorities among SUs, interrupt handling, timing & sequencing relationships, exception handling, concurrent execution, and other aspects of dynamic behavior, shall be included as applicable.

3.2.11. Interface design

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The following subparagraphs shall describe each of the interfaces among the SUs and their interfaces with internal and external interfaces, such as users, other SIs, HWCIs, and other systems.

3.2.11.1. Interface identification diagrams
Each interface description shall include:

3.2.11.1.1. Interface name.

3.2.11.1.2. Project unique interface identifier.

3.2.11.1.3.Purpose or type of interface.

3.2.11.1.4. Priority.

3.2.11.1.5. Identify the interfacing entity (SU, user, other SIs, HWCIs, and other systems) by its name, number, version, and documentation reference, as applicable.

3.2.11.1.6. State which entities have fixed interface characteristics (and therefore impose interface requirements on interfacing entities) and which are being developed or modified (thus having interface requirements imposed on them).

3.2.11.1.7. One or more interface diagram(s), as necessary, to depict the interface functionality.

3.2.11.1.8. Characteristics of the individual data elements provided by the interface, as applicable:

3.2.11.1.8.1. Non-technical data element name (descriptive name).

3.2.11.1.8.2. Project unique identifier .

3.2.11.1.8.3. Technical data element name (e.g. variable name).

3.2.11.1.8.4. Data type (e.g. string, integer, float, array, Boolean).

3.2.11.1.8.5. Size and format/structure.

3.2.11.1.8.6. Unit of measure (e.g. seconds, knots, m/s, feet, millibars, degrees).

3.2.11.1.8.7. Valid range (enumeration of possible values) (e.g. 00.0 – 99.9; “N, S, E, W”, True/False).

3.2.11.1.8.8. Accuracy (how correct).

3.2.11.1.8.9. Precision (number of significant digits).

3.2.11.1.8.10. Priority.

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3.2.11.1.8.11. Timing / frequency.

3.2.11.1.8.12. Bandwidth / data volume.

3.2.11.1.8.13. Source (setting or sending entity).

3.2.11.1.8.14. Destination (using or receiving entity).

3.2.11.1.8.15. Safety, cybersecurity, and privacy constraints.

3.2.11.1.9. Characteristics of the aggregate data assemblies, (e.g., displays, reports, messages, records, arrays, etc., as applicable:

3.2.11.1.9.1. Non-technical data assembly name.

3.2.11.1.9.2. Project unique identifier.

3.2.11.1.9.3. Data assembly technical name (e.g. record or data structure name used in the code or database).

3.2.11.1.9.4. Data elements and their structure/format in the data assembly.

3.2.11.1.9.5. Data storage medium (e.g. disk, memory).

3.2.11.1.9.6. Structure of data elements on or in the storage medium.

3.2.11.1.9.7. Visual and auditory characteristics of outputting data on displays, reports, messages, etc., such as colors, layouts, fonts, icons, tones, beeps, etc.

3.2.11.1.9.8. Relationships between and among other data assemblies.

3.2.11.1.9.9. Priority.

3.2.11.1.9.10. Timing / frequency.

3.2.11.1.9.11. Bandwidth / data volume.

3.2.11.1.9.12. Source (setting or sending entity).

3.2.11.1.9.13. Destination (using or receiving entity).

3.2.11.1.9.14. Safety, cybersecurity, and privacy constraints.

3.2.11.1.10. Characteristics of the interface communication methods, as applicable (Note: Publicly available commercial, and industry standard communication methods may be defined by reference to their documented standard):

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3.2.11.1.10.1. Project unique identifier.

3.2.11.1.10.2. Communication link characteristics.

3.2.11.1.10.3. Communication medium.

3.2.11.1.10.4. Message or data format.

3.2.11.1.10.5. Flow control mechanism.

3.2.11.1.10.6. Data transfer rate (e.g., bandwidth).

3.2.11.1.10.7. Synchronous or asynchronous.

3.2.11.1.10.8. Data transfer interval (e.g., period of transmission).

3.2.11.1.10.9. Data routing or addressing mechanism.

3.2.11.1.10.10. Safety, cybersecurity, and privacy constraints.

3.2.11.1.11. Characteristics of protocols used by the interface, as applicable, (Note: Publicly available commercial, and industry standard protocols may be defined by reference to their documented standard):

3.2.11.1.11.1. Project unique identifier.

3.2.11.1.11.2. Technical protocol name.

3.2.11.1.11.3. Protocol layer used.

3.2.11.1.11.4. Protocol priority.

3.2.11.1.11.5. Protocol standard.

3.2.11.1.12. Physical and other interface characteristics, as applicable:

3.2.11.1.12.1Connector / plug type.

3.2.11.1.12.2. Communication distance limitations.

3.2.11.1.12.3. Safety, cybersecurity, and privacy constraints.

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3.2.12. Detailed Design.

3.2.12.1. This section shall be divided into paragraphs that describe the detailed design of each SU comprising the SI for each SI.

3.2.12.2. This section shall include bidirectional traceability between the software requirements allocated to each SU, and those requirements allocated to the SI.

3.2.12.3. The detailed design of each SU shall contain the information necessary to construct the source code that performs the SUs functions as described.

3.2.12.4. Software unit, or group of units unique identifier.

3.2.12.4.1. This paragraph shall identify a SU by its project unique identifier and shall describe the detailed design of the unit.

3.2.12.4.2. Alternatively, this paragraph may identify a group of related SUs, and describe the detailed design of those SUs in subparagraphs. Naming conventions used to group and organize SUs shall be described.

3.2.12.4.3. Additional paragraphs and subparagraphs, as applicable, shall be created to describe the SUs.

3.2.12.4.4. Each SU detailed design description shall include the following information, as applicable:

3.2.12.4.4.1. Unit level design decisions: such algorithms and equations used; response times; handling of improper, un-allowed, and incorrect inputs or conditions; and other decisions similar to those applicable to SI-wide decisions.

3.2.12.4.4.2. Constraints, limitations, or unique capabilities and features of the SU design.

3.2.12.4.4.3. If the SU contains logic, describe the logic to be used by the SU, including the following:

3.2.12.4.4.3.1. Conditions in effect within the SU when execution is initiated.

3.2.12.4.4.3.2. Conditions under which control is passed to other SUs.

3.2..12.4.4.3.3. Responses (outputs) and response times to each input, including data conversion, renaming, and data transfer operations.

3.2.12.4.4.3.4. Sequence of operations and dynamically controlled sequencing during the SU's execution, including the method and logic of that control.

3.2.12.4.4.3.5. Input and output validation checking logic.

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3.2.12.4.4.3.6. Exception and error handling.

3.2.12.4.4.4. If the SU design contains, receives, or outputs data, a description of its inputs, outputs, other data elements, or aggregate data assemblies shall be provided.

3.2.12.4.4.5. If the SU design contains local data, describe that data, as applicable.

3.2.12.4.4.6. If the SU is a database, then a reference shall be provided to the interface characteristics or provide it here.

3.2.13. Additional Design Content
This section shall describe additional design content not included in other sections that were generated by specific and discrete task requirements as delineated in the contract. This section shall also include the derived design content not included in other sections that are not explicitly stated yet are required to satisfy one or more of the requirements.

3.2.14. Software Assurance
This section shall describe the merging of cybersecurity requirements into the software design and the decisions made for minimizing exposure to the vulnerabilities. This section shall describe the methods for handling the impact to the software design of emerging cybersecurity requirements.

3.2.15. Notes.

This section shall provide an alphabetic listing of acronyms, abbreviations, and their meaning as used in this SSPD list of terms and definitions needed to understand the SSPD or the appendices.

3.2.16. Appendixes

Appendixes may be used to provide information published separately for convenience in document maintenance (e.g., charts, classified data). As applicable, each appendix shall be referenced in the main body of the document where the data would normally have been provided. Appendixes may be bound as separate documents for ease in handling. Appendixes shall be lettered alphabetically (A, B, etc.)

End of DI-SESS-82466

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